

Seasonal Employment: Replication and Extension

Coglianesse & Price (2025) + QWI 6-Digit NAICS Index

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Outline

- 1 Replication of Coglianesse & Price (2025)
- 2 QWI 6-Digit NAICS Seasonal Index
- 3 Geographic Variation
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Coglianese & Price (2025): Overview

- “Income in the Off-Season: Household Adaptation to Yearly Work Interruptions,” *ILRR* 2025
- **Question:** How do households smooth income when employment separations recur annually (i.e. are *seasonal*)?
- **Key innovation:** Identify seasonal separations from administrative data without requiring industry or occupation labels
- Core estimand: **excess recurrence** at 12-month horizon

$$\widehat{\text{ExR}} = \text{Pr}(\text{sep at } k=12) - \frac{\text{Pr}(\text{sep at } k=11) + \text{Pr}(\text{sep at } k=13)}{2}$$

- Counterfactual = average of adjacent lags $\rightarrow$ isolates the *annual spike* marking genuine seasonal separations

Event-study OLS on separators panel (CPS and SIPP):

$$\mathbf{1}[\text{sep at } k] = \sum_j \beta_j \mathbf{1}[k = j] + \varepsilon$$

- Estimate by OLS; excess recurrence = $\beta_{12} - (\beta_{11} + \beta_{13})/2$
- Standard error via delta method
- **Two specifications:**
 - 1 *Raw* (Figure 2 in paper): only horizon dummies, no constant
 - 2 *Preferred* (Appendix F.1): adds controls for interview-seam artifacts (SIPP) and weeks elapsed since separation (CPS)
- Python replication uses `statsmodels` WLS with clustered SEs; data from the LEHD-style pre-cleaned `.dta.gz` files in the replication package

Replication Results

Table: Excess recurrence estimates: paper vs. Python replication

Specification	Where	Paper	Replication
CPS raw (Figure 2)	Figure 2	1.4 p.p. (SE 0.14)	1.44 ✓
CPS preferred	App. F.1	1.5 p.p. (SE 0.14)	1.42 ✓
SIPP raw (Figure 2)	Figure 2	2.0 p.p. (SE 0.11)	2.04 ✓
SIPP preferred	App. F.1	1.6 p.p. (SE 0.11)	1.62 ✓

- All four estimates within rounding of published values
- Industry ranking (both datasets): Agriculture > Entertainment > Education > Construction ≫ Healthcare, FIRE

Extending to 6-Digit NAICS via QWI

- CP's classifier operates at the worker level; industry rankings are reported at 1-digit NAICS only
- **Goal:** produce a finer, industry-level seasonal index suitable for firm classification
- **Data:** Census Quarterly Workforce Indicators (QWI)
 - Quarterly separations and employment by state $\times$ 6-digit NAICS
 - 51 states + DC, years 2000–2023 (1,011 industries)
- **Quarterly analog to excess recurrence:** for each quarter $q \in \{1, 2, 3, 4\}$ and industry j ,

$$\text{Excess}(q)_j = \text{SepRate}_{j,q,t} - \frac{\text{SepRate}_{j,q-1,t} + \text{SepRate}_{j,q+1,t}}{2}$$

(quarters treated cyclically; average over years t)

- **Peak excess:** $\text{ExcessPeak}_j = \max_q \text{Excess}(q)_j$
- **Seasonal amplitude:** $\max_q - \min_q$ normalised to $[0, 1]$; does not assume any particular quarter is “the” seasonal quarter

National Seasonal Index: Top Industries

Table: Top 12 most seasonal 6-digit NAICS industries (national)

NAICS & description	ExcessPeak	n
111336 Fruit/tree nut combo farm.	0.333	18
722330 Mobile food services	0.316	18
111140 Wheat farming	0.287	20
111160 Rice farming	0.561	22
111199 Other grain farming	0.397	23
111219 Other veg. (ex. potato) farm.	0.459	15
111998 Other misc. crop farming	0.328	20
111120 Oilseed (ex. soybean) farm.	0.347	12
115113 Crop harvesting (machine)	0.398	10
512132 Drive-in motion picture theaters	0.362	21
532284 Recreational goods rental	0.485	22
111191 Oilseed/grain combo farm.	0.321	23

- n = years contributing to ExcessPeak estimate; a ≥ 5 -year minimum is enforced so thin, disclosure-suppressed cells can't masquerade as extreme seasonality
- Non-agriculture entries are intuitive once named: mobile food services (food trucks), drive-in theaters, recreational goods rental
- Seasonal index normalized to $[0,1]$ via 99th-pct winsorization
- Correlation with CP 1-digit rankings: $\rho = 0.835$

National Seasonal Index: Sector Summary

Table: Mean seasonal index by 2-digit sector (national)

Sector	Mean seasonal index	ExcessPeak (p.p.)
Agriculture, forestry, fishing	0.471	15.0
Arts, entertainment & recreation	0.445	12.1
Accommodation & food services	0.323	8.3
Educational services	0.273	8.4
Construction	0.194	5.9
Real estate	0.167	5.6
Warehousing & couriers	0.135	4.6
Other services	0.134	4.2
Health care & social assistance	0.067	2.2
Wholesale trade	0.063	1.8
Manufacturing (metals/machinery)	0.036	1.1

Education peaks in Q2 (end of school year), not Q4 — ExcessPeak is positive across all sectors. CP's 1-digit rankings confirmed at finer granularity.

Robustness Check: Does COVID Distort the Index?

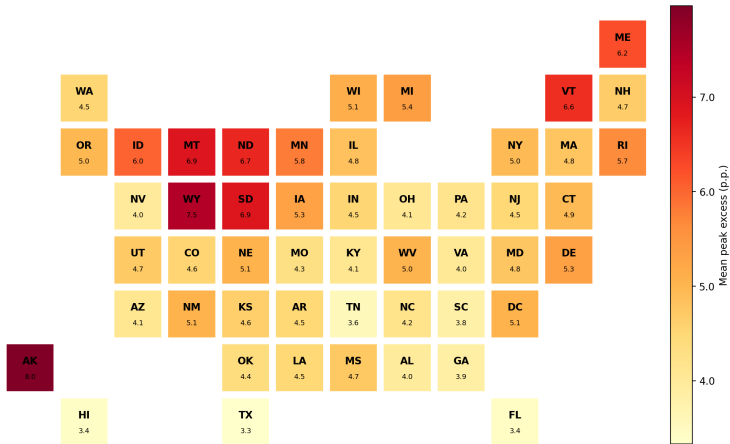
- The national index pools 2000–2023. Mass pandemic-era layoffs in industries already ranked highly seasonal (accommodation/food, arts & entertainment) could inflate “excess separation” for reasons that have nothing to do with normal annual seasonality
 - **Test:** recompute the index restricted to 2000–2019 only, compare to the full sample
 - **Result: rankings are robust.**
 - Pearson correlation (full vs. pre-COVID seasonal index): 0.997
 - Spearman rank correlation: 0.986
 - Only 6.3% of industries have a different peak quarter
 - Watch sectors (agriculture, construction, education, arts/entertainment, accommodation/food) all shift by ≤ 0.017
 - **One genuine mover:** a transportation industry (transit, NAICS 485111) whose index is roughly $5\times$ higher with COVID years included — plausibly a real ridership-collapse effect, not a data artifact
- Running this check is what surfaced a stale-data-threshold bug: three industries (incl. Tax Preparation Services, Cotton Ginning) had been wrongly ranked #1 nationally off just 2 years of COVID-distorted data, below the 5-year minimum — now fixed at the source

Geographic Variation: Overview

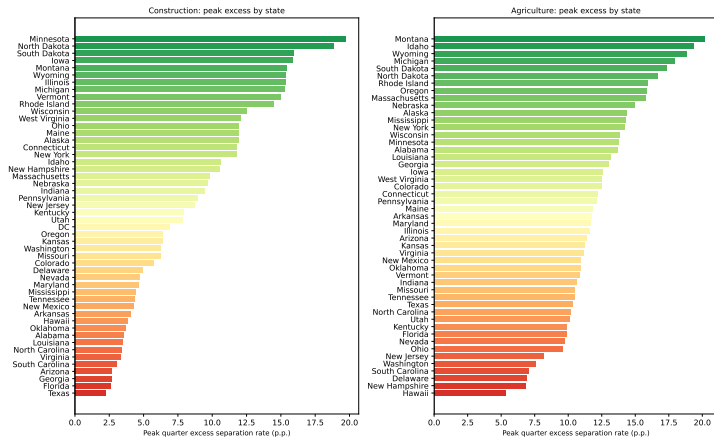
- The national index masks substantial state-level heterogeneity
- We compute ExcessPeak for each state $\times$ 6-digit NAICS cell (40,222 cells with ≥ 5 year-observations)
- **Main finding:** a $2.4\times$ gap between most and least seasonal states in mean peak excess (Alaska: 8.0 p.p. vs. Texas: 3.3 p.p.)
- Northern/Great Plains states consistently more seasonal across multiple sectors; Sun Belt states consistently less so
- **Most geographically variable sectors:**
 - 1 Agriculture (cross-state SD = 15.4 p.p.)
 - 2 Arts & entertainment (SD = 14.4 p.p.)
 - 3 Construction (SD = 9.9 p.p.)
- **Least geographically variable:** healthcare, metals manufacturing, utilities — consistent low seasonality everywhere

State-Level Mean Seasonality

State-level seasonality
Mean peak excess separation rate across 6-digit NAICS industries



Construction and Agriculture by State



Construction: Minnesota (19.7 p.p.) vs. Florida (2.6 p.p.) — an 8× gap driven by winter construction shutdowns in northern states. **Agriculture:** Montana (20.2 p.p.) and Idaho (19.3 p.p.) lead.

State-Specific Top and Bottom 1% Industries

Table: Within-state NAICS6 tails

Statistic	Top 1%	Bottom 1%
State-industry slots	413	413
Unique NAICS6 codes	133	252
Codes appearing once	57	158
Codes in 10+ states	6	0
Pairwise Jaccard	0.054	0.015
Also national same tail	10.9%	2.7%

Interpretation: the state-specific tails are related to the national index, but most state-tail industries are not simply the national top or bottom 1% repeated everywhere.

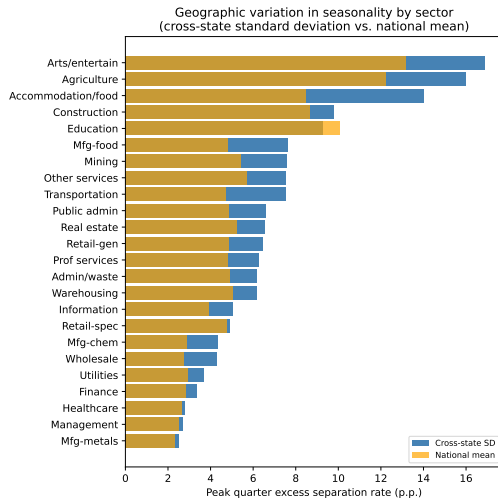
Recurring top-1% industries

- Nursery/tree production: 15 states
- Marinas: 14 states
- Specialty trade contractors: 12 states
- Golf courses/country clubs: 11 states
- Miscellaneous crop farming: 11 states

Bottom-1% pattern

- More diffuse than the top tail
- Concentrated in metals manufacturing, wholesale, chemicals, finance, and transportation

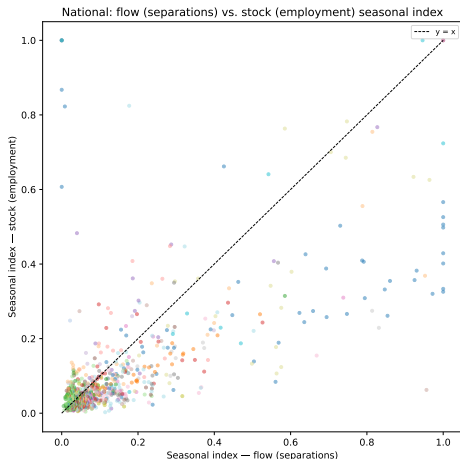
Geographic Variation: Sector Cross-State SD



Flow vs. Stock: Motivation

- The index has always been built two ways in parallel, but the two were never formally compared:
 - **Flow** (`seasonal_index`): from separation *rates* — does a quarter see an excess of separations?
 - **Stock** (`seasonal_index_emp`): from employment *levels* normalized to their own annual mean — does headcount itself swell or shrink that quarter?
 - **Result: correlated, but far from the same thing.**
 - National ($n = 1,004$): Pearson $r = 0.653$, Spearman $r = 0.627$
 - State $\times$ NAICS6 ($n = 39,225$): Pearson $r = 0.521$, Spearman $r = 0.585$
 - **Peak-quarter concordance is low — only $\sim 15\%$ nationally, $\sim 14\%$ by state:** even when both measures agree an industry is seasonal, they usually disagree on *which quarter* drives it
- Timing, not just amplitude, differs between “separations spike” and “headcount swings”

Flow vs. Stock: Sector Heterogeneity



Construction: near-perfect agreement

- Flow/stock correlation $r \approx 0.96$ – 0.97
- A direct, physical relationship: winter layoffs = winter headcount drop

Agriculture: the opposite

- High seasonal index on *both* measures, but $r \approx 0.21$ between them
- Consistent with rapid within-quarter worker replacement: departing workers are immediately replaced, spiking separations without moving headcount much

Validates the bug fix

- Tax Prep Services and Cotton Ginning — wrongly inflated on the flow side (previous slide) — now correctly show *low* flow, *high* stock: a real hiring-surge story, not an artifact

Implications for Firm Classification

- The QWI index enables firm-level seasonal classification via NAICS lookup
- The top/bottom 1% analysis shows the exact NAICS6 tails differ sharply across states: top tail has 133 unique codes; bottom tail has 252
- **But:** for geographically sensitive sectors (construction, agriculture), a single national index may misclassify:
 - A Minnesota construction firm is highly seasonal
 - A Florida construction firm is nearly non-seasonal
- **Preferred approach for firm classification:**
 - ① Use the *state* $\times$ *NAICS* ExcessPeak as the lookup value when state of operation is known
 - ② Fall back to national index for industries with thin state-level coverage (low *n_excess_obs*)
 - ③ Direct estimation from UI wage records when sufficient separations are available ($n \geq 20$ per firm)
- **Which index?** Flow suits classification by *turnover risk*; stock suits *headcount/staffing swings* — they agree only $\sim 65\%$ of the time nationally
- **Caveat:** 6-digit cells are often suppressed in small states; state-level estimates for niche agriculture sub-industries may rest on very few year-observations

Summary

- ➊ **Replication:** Python replication of CP (2025) excess recurrence matches all four published estimates within rounding
- ➋ **National index:** QWI-based ExcessPeak at 6-digit NAICS correlates strongly with CP's 1-digit rankings ($\rho = 0.835$); agriculture and entertainment top the seasonal hierarchy; robust to excluding 2020–2021 (Pearson 0.997)
- ➌ **Geographic variation:** large and systematic state heterogeneity — a $2.4\times$ gap between most and least seasonal states (Alaska vs. Texas); state-specific top/bottom 1% NAICS lists differ sharply from each other and from the national tails ($\sim 11\%$ overlap)
- ➍ **Flow vs. stock:** separations-based and employment-based seasonality are correlated but distinct ($r \approx 0.65$ national); peak-quarter concordance only $\sim 15\%$, with sharp sector heterogeneity (Construction: near-identical; Agriculture: not)
- ➎ **Next steps:** apply state $\times$ NAICS index to UI wage records for firm-level classification; county-level extension via CBP (validated, full run pending); income/earnings companion index (QWI refetch in progress)